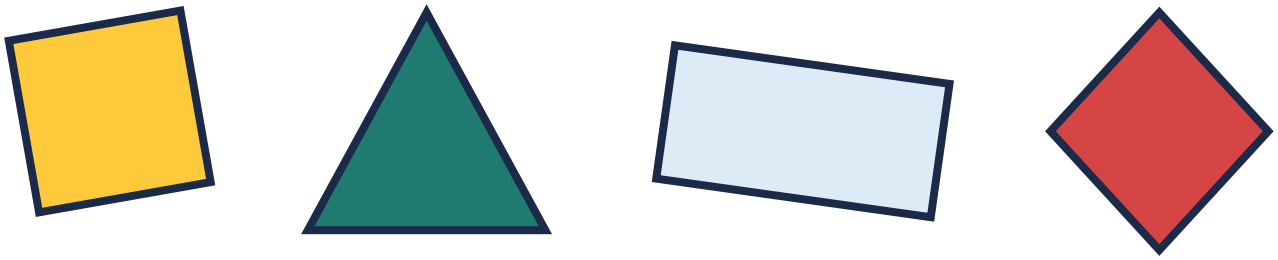
FOR PARENTS • PRIMARY 4 HOLIDAY PROGRAM

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

The three levels

Core is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Factor Hunters

LESSON 1 • FACTORS AND MULTIPLES

Core

1 Write all the factors of 12.

2 Write the first five multiples of 7.

3 Is 4 a factor of 52? How do you know?

4 Find a common multiple of 3 and 5 that is less than 20.

Think Harder

5 How many factors does 36 have? List them in order.

6 One lighthouse flashes every 6 seconds. Another flashes every 9 seconds. They flash together now. After how many seconds will they next flash together?

7 Find all the common factors of 24 and 36.

What your child learns

Factors and multiples: listing factor pairs in order, common factors and common multiples, and square numbers.

Tip for you

Use 24 coins or buttons. How many different equal rows can your child make? Which numbers have the most rows?

Ask

"How do you know you have found every factor?"

Challenge

8 How many numbers from 1 to 100 are multiples of 4 or multiples of 6 (or both)?

9 Mei has between 30 and 50 eggs. When she packs them in 4s, 1 egg is left. When she packs them in 5s, 1 egg is left. How many eggs does she have?

W1 Big Number Puzzles

LESSON 2 • NUMBERS TO 100 000

Core

1 In 64 205, what is the digit 4 worth?

2 Write fifty-three thousand and eighty in numerals.

3 Round 27 649 to the nearest thousand.

4 What number is 1000 less than 70 230?

Think Harder

5 What is the smallest whole number that rounds to 80 000 when rounded to the nearest thousand?

6 Use the digit cards 1, 4, 6, 7 and 9 once each. Make the 5-digit number that is closest to 70 000.

7 Find the rule and write the next two numbers: 17 250, 19 500, 21 750, ____, ____

What your child learns

Numbers to 100 000: rounding, estimating, and riddles about 5-digit numbers.

Tip for you

Ask your child to round the price of a car or flat in the news to the nearest thousand, then to the nearest ten thousand.

Ask

"Which digit decides how you round?"

Challenge

8 Ali counts on in 1000s, starting at 3456. He stops at the last number below 100 000. What is the last number he says? How many numbers does he say?

9 How many whole numbers round to 45 000 to the nearest thousand AND to 50 000 to the nearest ten thousand?

W2 LESSON 3 • FOUR OPERATIONS
Operation Detectives

Core

1 Work out 18 + 6 × 3.

2 Work out (18 + 6) × 3.

3 Work out 56 ÷ 7 × 2.

4 Work out 90 − 30 ÷ 5.

Think Harder

5 Work out 120 − 4 × (18 + 7).

6 Put ONE pair of brackets in 3 + 5 × 6 − 2 to make the answer 46.

7 Write +, −, × or ÷ in each gap to make this true: 12 __ 4 __ 5 = 15

What your child learns

Four operations: order of operations, brackets, missing signs and bar models.

Tip for you

Write 6 + 4 × 3 and ask for the answer. Then add brackets in different places and see how the answer changes.

Ask

"Which do you do first, and why?"

Challenge

8 Use four 4s and any of +, −, ×, ÷ and brackets to make 5.

9 Start at 7. If a number is even, halve it. If it is odd, multiply it by 3 and add 1. How many steps does it take to reach 1?

W2 Fraction Frontier

LESSON 4 • FRACTIONS

Core

1 Write 11/3 as a mixed number.

2 Write 2 3/4 as an improper fraction.

3 Find 1/4 of 32.

4 Find 2/3 of 18.

Think Harder

5 How many quarters are there in 4 3/4?

6 Siti has 40 beads. 3/8 of them are blue. How many beads are NOT blue?

7 Put these fractions in order, smallest first: 2/3, 1/2, 5/6, 7/12

What your child learns

Fractions: improper fractions and mixed numbers, comparing fractions, and fractions of a quantity with bar models.

Tip for you

Share a pizza or cake into equal slices. Ask for the number of slices as an improper fraction and as a mixed number.

Ask

"Can you show it with a bar?"

Challenge

8 How many fractions with denominator 12 are bigger than 1/4 but smaller than 2/3?

9 Mei spent 1/3 of her money on a book and 1/4 of her money on a pen. She had $25 left. How much money did she have at first?

W3 LESSON 5 • DECIMALS
Decimal Detectives

Core

1 Write 3 ones, 4 tenths and 7 thousandths as a decimal.

2 What is the value of the digit 6 in 12.365?

3 Which is greater, 0.9 or 0.89? How do you know?

4 Round 7.462 to 2 decimal places.

Think Harder

5 Write these in order, smallest first: 0.6, 0.066, 0.606, 0.66, 0.06

6 A number with 2 decimal places rounds to 10 when rounded to the nearest whole number. What is the smallest it can be? The largest?

7 What number is halfway between 2.6 and 2.7? What number is halfway between 2.6 and 2.61?

What your child learns

Decimals: ordering, rounding, and decimals in money and measures.

Tip for you

Look at a supermarket receipt. Order the prices from smallest to largest and round the total to the nearest dollar.

Ask

"What is each digit worth?"

Challenge

8 A number with 3 decimal places rounds to 2.46 when rounded to 2 decimal places. Its digits add up to 20. What is the number?

9 3 apples and 1 pear weigh 0.95 kg. 1 apple and 1 pear weigh 0.45 kg. All the apples weigh the same. What is the mass of the pear?

W3 LESSON 6 • ANGLES Angle Turns

Core

1 How many degrees are there in a half turn?

2 How many degrees are there in 3/4 of a turn?

3 Ali faces E. He turns 90° clockwise. Which direction does he face now?

4 Mei faces N. She turns 135° anticlockwise. Which direction does she face now?

Think Harder

5 Siti faces SE. She turns 225° clockwise. Which direction does she face now?

6 What is the angle between the clock hands at 10 o'clock? At 2 o'clock? What do you notice?

7 How many degrees does the minute hand turn from 7:15 to 7:50?

What your child learns

Angles: measuring with a protractor, estimating, and turns on a compass or clock.

Tip for you

Ask which angle the hands of a clock make at 4 o'clock, then check with a protractor.

Ask

"Is it more or less than a right angle?"

Challenge

8 What is the angle between the hands of a clock at 3:30?

9 Five rays come out of one point and make 4 equal angles side by side. The angle between the two outside rays is 120°. How many angles can you find altogether? How many of them are right angles?

W4 LESSON 7 • AREA AND PERIMETER
Area Architects II

Core

1 Find the area of a rectangle 9 cm long and 7 cm wide.

2 Find the perimeter of a rectangle 15 cm long and 4 cm wide.

3 A square has a perimeter of 36 cm. Find its area.

4 A rectangle has an area of 56 cm². Its length is 8 cm. Find its perimeter.

Think Harder

5 List every rectangle with whole-cm sides and an area of 30 cm². Which one has the smallest perimeter?

6 A rectangle has whole-cm sides and a perimeter of 20 cm. List all the possible rectangles and their areas. Which has the largest area?

What your child learns

Area and perimeter: figures made of rectangles, missing lengths, and same area with different perimeters.

Tip for you

Draw a room plan on squared paper. Ask for the area of the floor and the length of the skirting around it.

Ask

"Did you find all the missing lengths first?"

7 A 3 cm by 3 cm square is cut from each of the four corners of a 12 cm by 8 cm rectangle. Find the area and the perimeter of the shape that is left.

Challenge

8 A square is cut into two identical rectangles. Each rectangle has a perimeter of 24 cm. Find the area of the square.

9 A gap is cut into the middle of the top side of a 10 cm by 6 cm rectangle. The gap is 2 cm wide and 4 cm deep, with square corners. Find the area and the perimeter of the shape that is left.

W4 LESSON 8 • SYMMETRY AND NETS
Symmetry and Nets

Core

1 How many faces, edges and corners does a cube have?

2 How many lines of symmetry does a square have?

3 How many lines of symmetry does a rectangle that is not a square have?

4 On a dice, opposite faces add up to 7. Which number is opposite 2?

Think Harder

5 A cuboid is 3 cm long, 2 cm wide and 1 cm high. Its net is drawn on 1 cm squared paper. How many squares does the net cover?

6 Ben puts a dice on the table. The top face shows 3 and the front face shows 5. What numbers are on the bottom face and on the back face?

What your child learns

Symmetry and nets: finishing symmetric figures, lines of symmetry, and nets that fold into a cube.

Tip for you

Cut out a cross of six squares and see if it folds into a cube. Try other arrangements.

Ask

"Which faces are opposite each other?"

7 A dice is on a table. Mei walks round it and can see the top and the four side faces. The numbers she can see add up to 19. What number is on the bottom face?

Challenge

8 Two dice are glued together face to face, and the two glued faces show the same number. The numbers on all the faces you can still see add up to 36. What number is on the glued faces?

9 How many different nets of a cube are there? (Turned or flipped copies count as the same net.)

W5 Data Detectives

LESSON 9 • TABLES AND LINE GRAPHS

Core

1 On a line graph, each line on the scale is worth 10. A point is halfway between the 30 line and the 40 line. What number does it show?

2 A stall sold 45 bowls of laksa on Monday and 60 bowls on Tuesday. By how many bowls did sales go up?

3 A table shows 32 visitors on Monday, 48 on Tuesday and 20 on Wednesday. How many visitors came altogether?

4 A line graph goes up from 8 am to 10 am, then down until 12 pm. At what time was the number the highest?

Think Harder

5 Visitors to a museum: Monday 120, Tuesday 95, Wednesday 140, Thursday 110. Between which two days was the biggest change? Was it an increase or a decrease?

6 On a line graph each line on the scale is worth 20. A point is 3 lines above the 40 line. What number does it show?

What your child learns

Tables and line graphs: reading trends, finding missing values, and spotting misleading scales.

Tip for you

Track the temperature or your child's steps each day for a week and draw a line graph together.

Ask

"What does the steepest part tell us?"

7 A shop sold 300 kites in 4 days. Day 1: 65, Day 2: 80, Day 4: 90. How many kites were sold on Day 3?

Challenge

8 A shop's sales go up by the same amount every week. In week 2 it sold 45 toys and in week 6 it sold 85 toys. How many toys did it sell in week 1? In which week did it first sell more than 100 toys?

9 A stall sold 250 bowls of noodles from Monday to Friday. Monday 40 and Friday 50. Tuesday and Wednesday were the same, and Thursday was 10 more than Tuesday. Find the sales for Tuesday, Wednesday and Thursday.

W5 LESSON 10 • MIXED HEURISTICS Challenge Day

Core

1 Two next-door numbers (like 7 and 8) multiply to make 90. What are they?

2 A farm has chickens and rabbits. There are 8 heads and 22 legs. How many rabbits are there?

3 I think of a number. I add 15, then double it. I get 70. What is my number?

4 Ali is 6th from the front and 6th from the back of a queue. How many people are in the queue?

Think Harder

5 There are 20 bicycles and tricycles with 47 wheels altogether. How many tricycles are there?

6 Two next-door even numbers (like 6 and 8) multiply to make 624. What are they?

What your child learns

Choosing a strategy: guess and check, supposition, working backwards and logic grids.

Tip for you

When your child is stuck, ask "Can you draw it?" or "Can you try a number?" before giving any other hint.

Ask

"How did you start? What would you try next time?"

7 Ravi spent $15 on a book. Then he spent half of the money left on a bag. He had $18 left. How much money did he have at first?

Challenge

8 How many whole numbers from 1 to 100 have the digit 7 in them at least once?

9 A farm has 20 chickens and goats. The chickens have 4 more legs altogether than the goats. How many chickens are there?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Factor Hunters

1. 1, 2, 3, 4, 6, 12
2. 7, 14, 21, 28, 35
3. Yes: 4 × 13 = 52.
4. 15
5. 9 factors: 1, 2, 3, 4, 6, 9, 12, 18, 36.
6. 18 seconds (6, 12, 18 and 9, 18).
7. 1, 2, 3, 4, 6, 12
8. 33. Multiples of 4: 25. Multiples of 6: 16. Both (multiples of 12): 8, counted twice. 25 + 16 – 8 = 33.
9. 41. One less than the number of eggs is a common multiple of 4 and 5 (20, 40, 60), so 21, 41 or 61; only 41 is between 30 and 50.

Week 1 • Lesson 2: Big Number Puzzles

1. 4000 (4 thousands)
2. 53 080
3. 28 000
4. 69 230
5. 79 500
6. 69 741 (259 away; the closest number above 70 000 is 71 469, which is 1469 away).
7. Add 2250 each time: 24 000, 26 250.
8. 99 456. He says 97 numbers (3456, 4456, ... 99 456).
9. 500. To the nearest thousand: 44 500 to 45 499. To the nearest ten thousand: 45 000 to 54 999. Both: 45 000 to 45 499.

Week 2 • Lesson 3: Operation Detectives

1. 36
2. 72
3. 16
4. 84
5. 20 (18 + 7 = 25, 4 × 25 = 100, 120 – 100 = 20).
6. (3 + 5) × 6 – 2 = 8 × 6 – 2 = 46.
7. 12 ÷ 4 × 5 = 15 (the only way).
8. For example (4 × 4 + 4) ÷ 4 = 20 ÷ 4 = 5.
9. 16 steps: 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1.

Week 2 • Lesson 4: Fraction Frontier

1. 3 2/3
2. 11/4
3. 8
4. 12
5. 19 (4 wholes = 16 quarters, plus 3).
6. 25 (1 unit = 5, blue = 15, not blue = 5 units = 25).
7. 1/2, 7/12, 2/3, 5/6 (6/12, 7/12, 8/12, 10/12).
8. 4: 4/12, 5/12, 6/12, 7/12 (1/4 = 3/12 and 2/3 = 8/12 do not count).
9. $60. Book 4/12, pen 3/12, so 5/12 is left. 5 units = $25, 1 unit = $5, 12 units = $60.

Week 3 • Lesson 5: Decimal Detectives

1. 3.407
2. 6 hundredths (0.06)
3. 0.9. Line up the points: 0.90 and 0.89. 9 tenths is more than 8 tenths.
4. 7.46
5. 0.06, 0.066, 0.6, 0.606, 0.66 (as 0.060, 0.066, 0.600, 0.606, 0.660)
6. Smallest 9.50, largest 10.49
7. 2.65 and 2.605
8. 2.459 (it is from 2.455 to 2.464; 2.45_ needs 9 to make 20; 2.46_ would need 8, but 2.468 rounds to 2.47)
9. 0.2 kg (2 apples = 0.95 − 0.45 = 0.5 kg, 1 apple = 0.25 kg, pear = 0.45 − 0.25)

Week 3 • Lesson 6: Angle Turns

1. 180°
2. 270°
3. S
4. SW
5. N (225° is 5 steps of 45°: S, SW, W, NW, N)
6. 60° and 60°. Both are 2 spaces apart the shorter way.
7. 210° (35 minutes × 6°)
8. 75° (the minute hand points to 6; the hour hand is halfway from 3 to 4, so it is 2 and a half spaces from the 6: 30° + 30° + 15°)
9. 10 angles (4 + 3 + 2 + 1). Each small angle is 30°, so the right angles are the 3-part angles: 2 of them.

Week 4 • Lesson 7: Area Architects II

1. 63 cm²
2. 38 cm
3. 81 cm² (each side is 9 cm).
4. 30 cm (the breadth is 7 cm; 8 + 7 + 8 + 7 = 30).
5. 1 by 30 (62 cm), 2 by 15 (34 cm), 3 by 10 (26 cm), 5 by 6 (22 cm). Smallest: 5 cm by 6 cm, 22 cm.
6. 1 by 9 (9 cm²), 2 by 8 (16 cm²), 3 by 7 (21 cm²), 4 by 6 (24 cm²), 5 by 5 (25 cm²). Largest: the 5 cm by 5 cm square, 25 cm².
7. Area 96 − 4 × 9 = 60 cm². Perimeter 40 cm, the same as the rectangle: each cut takes away 3 + 3 and adds 3 + 3.
8. 64 cm². Each rectangle is one whole side and one half side all round, which is 3 sides of the square. 24 ÷ 3 = 8 cm, and 8 × 8 = 64.
9. Area 60 − 8 = 52 cm². Perimeter 32 + 4 + 4 = 40 cm: the two 4 cm sides inside the gap cannot be pushed out.

Week 4 • Lesson 8: Symmetry and Nets

1. 6 faces, 12 edges, 8 corners.
2. 4
3. 2 (the diagonals are not lines of symmetry).
4. 5
5. 22 squares: 2 faces of 3 by 2 (6 each), 2 faces of 3 by 1 (3 each) and 2 faces of 2 by 1 (2 each). 12 + 6 + 4 = 22.
6. Bottom 4 (3 + 4 = 7). Back 2 (5 + 2 = 7).
7. 2. All six faces add up to 1 + 2 + 3 + 4 + 5 + 6 = 21, and 21 − 19 = 2.
8. 3. Two dice have 21 + 21 = 42 altogether. The hidden faces add up to 42 − 36 = 6, and they show the same number, so each is 3.
9. 11.

Week 5 • Lesson 9: Data Detectives

1. 35
2. 15 bowls
3. 100
4. 10 am
5. Tuesday to Wednesday: an increase of 45. (The other changes are down 25 and down 30.)
6. 100 (40 + 20 + 20 + 20)
7. 65 (300 – 65 – 80 – 90)
8. Week 1: 35 (4 jumps make 40, so 10 a week). First more than 100: week 8 (105).
9. Tuesday 50, Wednesday 50, Thursday 60. (Tue + Wed + Thu = 160; take away the extra 10, 150 shared by 3 = 50.)

Week 5 • Lesson 10: Challenge Day

1. 9 and 10
2. 3 rabbits (and 5 chickens). Suppose all chickens: 16 legs, 6 short, each swap adds 2.
3. 20 (70 ÷ 2 = 35, 35 – 15 = 20)
4. 11 (5 + 1 + 5)
5. 7 tricycles (and 13 bicycles). All bicycles: 40 wheels, 7 short, each swap adds 1.
6. 24 and 26
7. $51 (18 × 2 = 36, 36 + 15 = 51)
8. 19 (7, 17, 27, 37, 47, 57, 67, 87, 97 and 70 to 79; 77 is counted once)
9. 14 chickens (28 legs) and 6 goats (24 legs).